PRACH Preamble Design for 6G Wireless Systems

Fan Zhang[1], Hanyu Liu[1], Zihan Liu[1], Tianqi Mao[2], *Member, IEEE*,
Christos Masouros[3], *Fellow, IEEE*, and Zhaocheng Wang[1], *Fellow, IEEE*,
[1]Department of Electronic Engineering, Tsinghua University, Beijing 100080, China
[2]State Key Laboratory of Environment Characteristics and Effects for Near-space,
Beijing Institute of Technology, Beijing 100081, China
[3] Department Electronics & Electrical Engineering, University College London, London WC1E 7JE, UK

***Abstract*—Motivated by cutting-edge applications such as the massive Internet of Things, next-generation wireless networks are expected to accommodate millions of devices per square kilometer. Such massive access places rigorous demands on the design of physical random access channel preambles, which are required to possess high capacity, Doppler robustness, and ideal correlation properties. However, existing sequences that meet these criteria are usually hindered by prohibitive detection complexity. To tackle this challenge, a Kronecker-product structured (KPS) sequence is proposed that naturally embeds a multi-segment structure, enabling the low-complexity detection. Furthermore, a two-stage low-complexity detection algorithm is proposed, which decouples inner and outer correlations and simplifies the Doppler compensation process in order to substantially reduce the computational overhead. Simulation results demonstrate that employing KPS sequences with the proposed detection algorithm achieves accurate detection while maintaining low overhead.**

***Index Terms*—Physical random access channel (PRACH), waveform design, orthogonal frequency division multiplexing (OFDM), ambiguity function**

I. Introduction

The rapid evolution of wireless terminals and data-hungry services has triggered an unprecedented demand for massive connectivity in the next-generation networks [1], [2]. From large-scale sensor deployments in smart cities to high-mobility communication in vehicular networks, supporting a huge number of devices has become a critical objective for sixth-generation (6G) networks [3]. Such large-scale connectivity imposes substantial challenges for random access procedures, particularly in the design of the underlying waveform and sequence set [4].

In current networks, users (UEs) initiate connection through the physical random access channel (PRACH) [5]. Distinct PRACH signals, known as preambles, are allocated to UEs and transmitted for random access. These preambles allow the base station (BS) to identify active UEs and estimate their timing offsets [6]. To achieve reliable detection, preambles are generally selected from sequence sets characterized by preferable auto- and cross-correlation properties.

The Zadoff-Chu (ZC) sequence set, constructed using distinct roots and cyclic shifts, has been extensively adopted as a standard choice for PRACH preambles due to its ideal auto-correlation and low cross-correlation between sequences of different roots [7]. Nevertheless, in high-mobility scenarios, severe Doppler shifts distort its perfect auto-correlation property [8]. To address this, the restricted set was introduced to eliminate the Doppler-induced sidelobe ambiguities, but this approach drastically limits the number of available preambles [9]. This drawback reveals that traditional ZC sequences can no longer satisfy the requirements of massive access.

To enable massive access under high-mobility conditions, recent studies on PRACH preambles have sought to enlarge the sequence set cardinality and enhance Doppler robustness. ZCm and ZCa sequences in [10] expand the PRACH capacity by element-wise multiplying ZC sequences with M-sequences [11] and Alltop sequences [12], respectively. Such masking alters the original quadratic phase structure of the ZC sequence, which is known to be highly Doppler-sensitive, thereby improving their robustness to frequency offsets. In [13], a different strategy was proposed, where two ZC sequences with distinct roots are aggregated to form a larger set by enumerating all possible sequence pairs. Although existing designs can increase the preamble set cardinality, they fail to address the consequent rise in detection complexity. Since the receiver is required to compute the correlation between the received signal and every sequence in the preamble set separately and independently, the number of required multiplications scales linearly with the sequence capacity, imposing substantial computational load on the BS

In response to these challenges, we investigate a novel high-capacity PRACH sequence design methodology, aiming to obtain a sequence with specific structure characteristics that can be leveraged to simplify the detection algorithm through computational reuse. Specifically, we develop a Kronecker-product structured (KPS) sequence family, which is constructed by performing the Kronecker product on two distinct types of sequences. Furthermore, we propose a low-complexity detection algorithm based on the structure of the proposed sequences, which breaks the proportional relationship between PRACH capacity and complexity. Numerical results demonstrate that employing KPS sequences with the proposed detection algorithm can achieve large-scale connectivity with accurate detection and low overhead.

Notation: The operator $\log(\cdot)$ denotes the base-2 logarithm. The notation $\langle\cdot\rangle_N$ represents the modulo operation with divisor N, and $|\cdot|$ refers to the modulus. The floor operator $\lfloor\cdot\rfloor$ rounds a real number to the nearest integer less than or equal to that number. The superscript $(\cdot)^*$ indicates the conjugate operation. An N-length sequence is written as $\mathbf{x} = (x[0], x[1], \cdots, x[N-1])$, where $x[n]$ denotes the n-th element of the sequence.

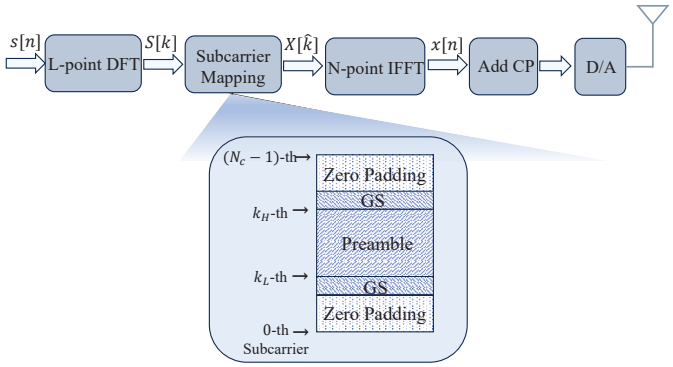


Fig. 1. PRACH transmitter.

II. System Model

A. PRACH Transmitter

To guarantee uninterrupted data transmission, each UE is required to establish a connection with the BS prior to transmitting data packets. Consider a preamble set denoted as $\mathcal{P} = \{\mathbf{s}_0, \mathbf{s}_1, \cdots, \mathbf{s}_{N-1}\}$, where N represents the set cardinality and L denotes the length of each sequence. Each UE selects one sequence from $\mathcal{P}$ and applies a distinct cyclic shift to generate its unique preamble. We assume that the p-th UE occupies the p-th sequence $\mathbf{s}_p$ as its preamble. For brevity, we denote $\mathbf{s}_p$ as $\mathbf{s}$ by omitting its subscript. The preamble transmission is generally based on the discrete Fourier transform-spread orthogonal frequency division multiplexing (DFT-s-OFDM) waveform. As illustrated in Fig. 1, the sequence $\mathbf{s}$ is transformed into the frequency domain by performing the L-point discrete Fourier transform (DFT), which can be expressed as

$$S[k] = \frac{1}{\sqrt{L}} \sum_{n=0}^{L-1} s[n] e^{-\mathrm{j}2\pi \frac{nk}{L}}, \ k = 0, 1, \cdots, L-1. \quad (1)$$

Then, the frequency-domain sequence $S[k]$ is mapped onto continuous L subcarriers among the N_c total OFDM subcarriers. After subcarrier mapping, the guard spacing (GS) is added at both the top and bottom of the preamble in the frequency domain. When no other signals need to be carried within an OFDM symbol, the remaining subcarriers are padded with zeros to complete the symbol. For simplicity, the preamble sequence is assumed to be mapped onto the central L subcarriers of the OFDM symbol, and the GS is filled with zeros. Consequently, the signal in each subcarrier of the transmitted OFDM symbol $\mathbf{X}$ can be formulated as

$$X[\hat{k}] = \begin{cases} S\left[\hat{k} - \left\lfloor \frac{N_c - L}{2} \right\rfloor\right], \left\lfloor \frac{N_c - L}{2} \right\rfloor \le \hat{k} < \left\lfloor \frac{N_c + L}{2} \right\rfloor \\ 0, \ \text{otherwise}, \end{cases} \quad (2)$$

where $X[\hat{k}]$ denotes the signal in the $\hat{k}$-th subcarrier with $\hat{k} = 0, 1, \cdots, N_c - 1$. For brevity, we denote $\left\lfloor \frac{N_c - L}{2} \right\rfloor$ and $\left\lfloor \frac{N_c + L}{2} \right\rfloor$ as k_L and k_H in Fig. 1, respectively. After performing the inverse discrete Fourier transform (IDFT) on $X[\hat{k}]$, the OFDM symbol in the time domain can be expressed as

$$x[n] = \frac{1}{\sqrt{N_c}} \sum_{\hat{k}=0}^{N_c-1} X[\hat{k}] e^{\mathrm{j}2\pi \frac{n\hat{k}}{N_c}}, \ n = 0, 1, \cdots, N_c - 1. \quad (3)$$

To mitigate the inter-symbol interference (ISI), a cyclic prefix (CP) of length N_{p} is inserted before each OFDM symbol. After adding the CP, the discrete signals are fed into a digital-to-analog converter (D/A) to obtain the continuous transmit signal as

$$x(t) = \frac{1}{\sqrt{N_c}} \sum_{\hat{k}=0}^{N_c-1} X[\hat{k}] e^{\mathrm{j}2\pi \frac{\hat{k}(t - N_{\mathrm{p}} T_s)}{N_c T_s}}, \quad (4)$$

where $t \in \left[0, \ (N_c + N_{\mathrm{p}}) T_s\right)$ and T_s is the sampling period.

B. PRACH receiver

Assuming the BS is equipped with M antennas and each UE is equipped with a single antenna, the clustered delay line (CDL) single-input multiple-output (SIMO) fading channel is considered for each UE [14]. Let $g^{(m)}(\tau, t)$ denote the channel impulse response of the m-th receiver antenna. The received signal at the m-th antenna can be expressed as

$$y^{(m)}(t) = \int_0^{\infty} g^{(m)}(\tau, t) x(t - \tau) d\tau + w^{(m)}(t), \quad (5)$$

where $w^{(m)}(t) \sim \mathcal{CN}(0, \sigma_{\mathrm{c}}^2)$ represents the Gaussian white noise. Following analog-to-digital (A/D) conversion and CP removal, the discrete received signal can be represented as

$$y^{(m)}[n] = y^{(m)}(nT_s), \ n = 0, 1, \cdots, N_c - 1. \quad (6)$$

Upon receiving the signal, the BS identifies the UE and estimates its timing offset via the detection algorithm. To illustrate the detection process, we introduce a conventional algorithm [15], as presented in Fig. 2. Suppose that the normalized Doppler frequency shift range is $[-V, V]$, where V represents the maximum normalized Doppler frequency shift. To combat the Doppler shift, the received signals undergo parallel frequency offset compensation. Let $\Delta\nu$ denote the resolution of frequency offset compensation. Then the number of Doppler compensation points can be written as $N_{\mathrm{dop}} = \frac{2V}{\Delta\nu} + 1$. The received signal at the m-th antenna compensated by the μ-th frequency point can be expressed as

$$\hat{y}^{(m,\mu)}[n] = y^{(m)}[n] e^{-\mathrm{j}2\pi n \frac{\mu\Delta\nu - V}{N_c}}, \quad (7)$$

where $\mu = 0, 1, \cdots, N_{\mathrm{dop}} - 1$.

For each Doppler-compensated version of the signal, an N_c-point FFT is performed to convert the signal into the frequency domain, from which the preamble is extracted according to the preassigned subcarrier indices. We denote the obtained frequency-domain preamble as $Y^{(m,\mu)}[k], \ k = 0, 1, \cdots, L$–1. Subsequently, the compensated signals are correlated with each sequence in $\mathcal{P}$ to obtain the correlation matrices. The correlation is typically carried out via frequency-domain multiplication followed by transformation to the time domain,

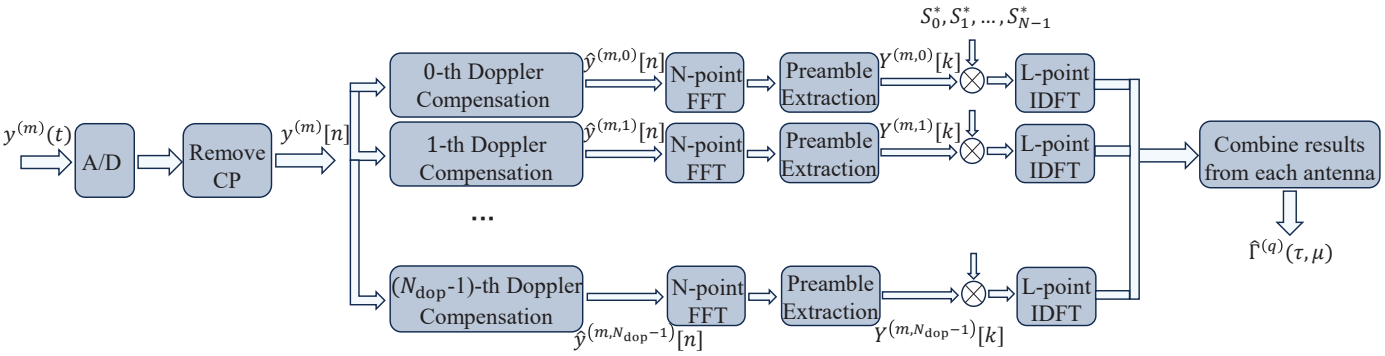


Fig. 2. PRACH receiver adopting the conventional detection algorithm.

which significantly reduces the computational complexity. Let $\mathbf{\Gamma}^{(m,q)} \in \mathbb{C}^{N_c \times N_{\text{dop}}}$ represent the correlation results with the q-th sequence $\mathbf{s}_q$ in $\mathcal{P}$, which can be written as

$$\Gamma^{(m,q)}(\tau,\mu) = \sum_{k=0}^{L-1} S_q^*[k] Y^{(m,\mu)}[k] e^{\mathrm{j}2\pi \frac{\tau k}{L}}, \tag{8}$$

where $\mathbf{S}_q$ denotes the frequency-domain expression of $\mathbf{s}_q$ and $S_q[k]$ is the k-th element of $\mathbf{S}_q$. Here τ and μ are the indices in the time and Doppler domains, respectively. To aggregate the energy of the main peak, we compute the element-wise sum of the magnitudes of the correlation matrices corresponding to each antenna to obtain the final decision matrix $\hat{\mathbf{\Gamma}}^{(q)}$ as

$$\hat{\Gamma}^{(q)}(\tau,\mu) = \sum_{m=0}^{M-1} \left| \Gamma^{(m,q)}(\tau,\mu) \right|. \tag{9}$$

Given the N matrices $\hat{\mathbf{\Gamma}}^{(0)}, \hat{\mathbf{\Gamma}}^{(1)}, \cdots, \hat{\mathbf{\Gamma}}^{(N-1)}$, the target can be detected via the hypothesis test, allowing the BS to deduce the UE's identity and its timing offset.

III. Proposed KPS Sequence

To overcome the limitations of PRACH capacity, we propose a sequence design methodology based on the Kronecker product to derive the KPS sequence, which is constructed as the Kronecker product of two component sequences. Let $\mathcal{P}_O = \{\mathbf{p}_0, \mathbf{p}_1, \cdots, \mathbf{p}_{N_O-1}\}$ and $\mathcal{P}_I = \{\mathbf{q}_0, \mathbf{q}_1, \cdots, \mathbf{q}_{N_I-1}\}$ denote the outer and inner sequence sets, respectively. Suppose that the lengths of each outer and inner sequence are L_O and L_I, respectively. The proposed KPS sequence set can be constructed as

$$\mathcal{P} = \{\mathbf{p}_i \otimes \mathbf{q}_j | \mathbf{p}_i \in \mathcal{P}_O, \mathbf{q}_j \in \mathcal{P}_I\}, \tag{10}$$

where $\otimes$ denotes the Kronecker product operation and the cardinality of $\mathcal{P}$ is $N = N_O N_I$. Specifically, the resultant KPS sequence can be expressed as

$$\mathbf{p}_i \otimes \mathbf{q}_j = (p_i[0]\mathbf{q}_j, p_i[1]\mathbf{q}_j, \cdots, p_i[L_O-1]\mathbf{q}_j), \tag{11}$$

which can be conceptually viewed as multiple segments of the inner sequence, each scaled by a different element of the outer sequence and then concatenated, as illustrated in Fig. 3. We denote $\mathbf{p}_i \otimes \mathbf{q}_j$ as the $(iN_I + j)$-th sequence in $\mathcal{P}$, written as

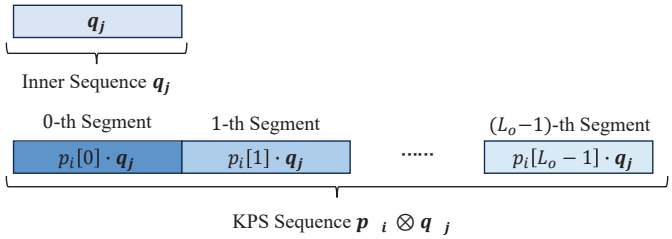


Fig. 3. The structure of KPS sequence.

$\mathbf{s}_{(iN_I+j)}$. The length of $\mathbf{s}_{(iN_I+j)}$ is $L = L_O L_I$ and its element can be represented as

$$s_{(iN_I+j)}[n_1 L_I + n_2] = p_i[n_1]\, q_j[n_2]. \tag{12}$$

where $n_1 = 0, 1, \cdots, L_O - 1$ and $n_2 = 0, 1, \cdots, L_I - 1$.

The UE detection is typically performed within a limited delay range (not exceeding N_{cs}). Therefore, different cyclically shifted versions of the same KPS sequence can be assigned to different UEs, and each preamble can be generated by

$$s_{(iN_I+j,\delta)}[n] = s_{(iN_I+j)}[\langle n - \delta N_{\text{cs}} \rangle_L] \tag{13}$$

where $\delta \in \left\{0, 1, \cdots, \left\lfloor \frac{L}{N_{\text{cs}}} \right\rfloor - 1\right\}$. The KPS sequence set $\mathcal{P}$ can support up to $N_O N_I \left\lfloor \frac{L}{N_{\text{cs}}} \right\rfloor$ UEs. To ensure desirable properties of the resulting KPS sequence set, guidelines on the selection of the inner and outer sequences are outlined as follows.

1) Both inner and outer sequence sets should consist of sequences with ideal auto- and cross-correlation values to ensure favorable correlation properties of the KPS sequence.
2) The outer sequence should exhibit strong Doppler robustness, whereas the Doppler tolerance requirement for the inner sequence can be relatively relaxed, since the effective Doppler shift it experiences is reduced by a factor of $1/L_O$.
3) When the outer sequence is binary, the proposed detection algorithm can achieve a further reduction in computational complexity. The detailed detection process will be introduced in Section IV.

In brief, the outer sequence can be selected from binary sequences exhibiting good correlation and strong Doppler

robustness, such as M-sequences and Gold sequences, whereas the inner sequence can be chosen from constant-modulus sequences with desirable correlation characteristics, such as ZC and Alltop sequences.

In this paper, we elaborate on two representatives in KPS sequence family, which are the m-ZC KPS and m-Alltop KPS sequences. They are obtained by taking the Kronecker product of M-sequences and ZC/Alltop sequences. The i-th sequence in the M-sequence set of length $L_O = 2^r - 1$ is defined as

$$p_i[n] = (-1)^{\mathrm{Tr}(\alpha_i^n)},\ n = 0, 1, \cdots, L_O - 1, \tag{14}$$

where $\mathrm{Tr}(x) = \sum_{\beta=0}^{r-1} x^{2^\beta}$ is the trace function from Galois field extension $\mathrm{GF}(2^r)$ to $\mathrm{GF}(2)$ and α_i is a primitive element of $\mathrm{GF}(2^r)$. According to the number of primitive elements, the capacity of M-sequence set can be expressed as $N_O = \varphi(L_O)/\log_2(L_O + 1)$, where $\varphi(L_O)$ is the Euler function to obtain the number of positive integers less than or equal to L_O that are coprime to L_O. The ZC sequence can be obtained by

$$q_u[n] = e^{-\mathrm{j}\frac{\pi u n(n+1)}{L_I}},\ n = 0, 1, \cdots, L_I - 1, \tag{15}$$

where the length L_I is a prime number and the root index u satisfies $u \in \{1, \cdots, L_I - 1\}$. The capacity of ZC sequence set is $P_I = L_I - 1$. Therefore, the m-ZC KPS sequence set, generated by taking the Kronecker product between all pairs of M-sequences and ZC sequences, has a total capacity of

$$N = (L_I - 1)\varphi(L_O)/\log_2(L_O + 1). \tag{16}$$

Additionally, an Alltop sequence can be formulated as

$$q_{(\lambda,\omega)}[n] = e^{-\mathrm{j}2\pi\frac{(n+\omega)^3+\lambda n}{L_I}},\ n = 0, 1, \cdots, L_I - 1, \tag{17}$$

where λ and ω are integers satisfying $0 \le \lambda, \omega \le L_I - 1$. The L_I-length Alltop sequence set contains a total of L_I^2 sequences. Accordingly, the capacity of m-Alltop KPS sequence set can be expressed as

$$N = L_I^2\varphi(L_O)/\log_2(L_O + 1). \tag{18}$$

IV. Proposed Two-Stage Low-Complexity Detection Algorithm

Leveraging a built-in segment-structure property, the proposed KPS sequence family surpasses conventional designs by enabling the separation of correlation calculations in detection into inner-layer and outer-layer correlation. The partitioned correlation computations substantially increases the reuse of intermediate outputs. Additionally, due to the significantly lower effective Doppler shift experienced by the inner sequences, Doppler compensation is only necessary for outer sequences, which further reduces the overall computation. To capitalize on these characteristics, a two-stage detection scheme with low computational overhead is introduced. In the first stage, a simplified cross-correlation and Doppler compensation process are performed to efficiently obtain a set of N_m candidate UEs. Subsequently, these N_m potential targets undergo refined detection in the second stage to achieve higher detection accuracy.

Without loss of generality, the proposed method is presented under a single-antenna setup, as the received signals at each antenna undergo identical and independent processing. By summing the individual correlation matrices corresponding to all antennas to obtain the aggregate correlation matrix, the proposed approach can be readily generalized to a multi-antenna system.

***Stage I: Low-complexity Doppler Compensation**:* Suppose that the frequency-domain received signal in the k-th subcarrier based on the DFT-s-OFDM technology is $Y[k]$ with $k = 0, 1, \cdots, N_c - 1$. The frequency-domain representation of the received preamble is extracted from L preallocated subcarriers, i.e., $\hat{Y}[k] = Y\left[k + \left\lfloor\frac{N_c - L}{2}\right\rfloor\right], k = 0, 1, \cdots, L - 1$, following Section II. To derive the inner-layer delay shift, the cross-correlation between the received signal and each inner sequence in $\mathcal{P}_I$ is calculated. Specifically, with inner sequences transferred into the frequency domain, the correlation can be efficiently computed via element-wise multiplication in the frequency domain and inversely transformation back to the time domain. Based on L-point DFT, the frequency-domain representation of the j-th inner sequence in $\mathcal{P}_I$, expressed as $Q_j[k] = \sum_{n=0}^{L_I-1} q_j[n]e^{-\mathrm{j}2\pi\frac{nk}{L}}$, is precomputed and stored at the BS, thus introduces negligible computational overhead. Then the cross-correlation result can be written as

$$c_j[\tau] = \sum_{k=0}^{L-1} Q_j[k]\hat{Y}[k]e^{\mathrm{j}2\pi\frac{\tau k}{L}},\ \tau = 0, \cdots, L - 1. \tag{19}$$

As the power-of-two-point DFT can be efficiently implemented utilizing the FFT algorithm, an N_c-point FFT is adopted for each inner sequence. Subsequently, the L-point result c_j in (19) can be obtained by downsampling the N_c-point result of FFT. Therefore, the number of multiplications required for each inner sequence is $N_c \log N_c + L$, and the total number of multiplications amounts to $N_I\left(N_c \log N_c + L\right)$. Each cross-correlation result $c_j \in \mathbb{C}^L$ is reshaped into a matrix $\mathbf{C}_j \in \mathbb{C}^{L_O \times L_I}$, whose (τ_1, τ_2)-th element is given by

$$C_j[\tau_1, \tau_2] = c_j[\tau_1 L_I + \tau_2], \tag{20}$$

with $\tau_1 = 0, \cdots, L_O - 1$ and $\tau_2 = 0, \cdots, L_I - 1$.

The cross correlation between each column of $\mathbf{C}_j$ and each outer sequence in $\mathcal{P}_O$ is subsequently calculated, with the Doppler shift for the outer-layer compensated simultaneously, which can be formulated as

$$D_{i,j}[\nu, \tau_2, \tau_3] = \sum_{\tau_1=0}^{L_O-1} e^{-\mathrm{j}2\pi\frac{\tau_1\nu}{N_{\mathrm{dop}}}} p_i[\langle\tau_1 - \tau_3\rangle_{L_O}]C_j[\tau_1, \tau_2] \tag{21}$$

with $\tau_3 = 0, 1, \cdots, L_O - 1$ and $\nu = 0, 1, \cdots, N_{\mathrm{dop}} - 1$. We denote the calculation result of $p_i[\langle\tau_1 - \tau_3\rangle_{L_O}]C_j[\tau_1, \tau_2]$ as $\hat{C}_{i,j}[\tau_1, \tau_2, \tau_3]$. When outer sequences are binary, i.e., $p_i[\langle\tau_1 - \tau_3\rangle_{L_O}] \in \{-1, 1\}$, calculating $\hat{C}_{i,j}[\tau_1, \tau_2, \tau_3]$ requires no extra multiplications. Then (21) can be further simplified as

$$D_{i,j}[\nu, \tau_2, \tau_3] = \sum_{\tau_1=0}^{L_O-1} e^{-\mathrm{j}2\pi\frac{\tau_1\nu}{N_{\mathrm{dop}}}} \hat{C}_{i,j}[\tau_1, \tau_2, \tau_3]. \tag{22}$$

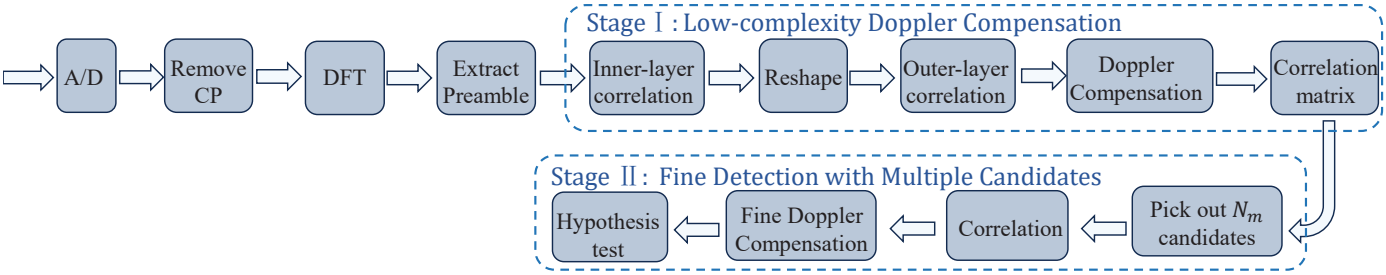


Fig. 4. PRACH receiver adopting the proposed two-stage low-complexity detection algorithm.

with $i = 0, \cdots, N_O - 1$ and $j = 0, \cdots, N_I - 1$. Noting that for fixed τ_2, τ_3, i and j, (22) is equivalent to an N_{dop}-point DFT along the first dimension of matrix $\hat{\mathbf{C}}_{i,j} \in \mathbb{C}^{L_O \times L_I \times L_O}$, FFT can be utilized to significantly reduce the time of multiplications. Supposing that the number of FFT points, which decides the resolution of Doppler shift estimation, is set to N_{dop}, $NLN_{\text{dop}} \log N_{\text{dop}}$ times of multiplication is required to obtain all the correlation matrices $\mathbf{D}_{i,j} \in \mathbb{C}^{N_{\text{dop}} \times L_I \times L_O}$. Among all NLN_{dop} elements from all the correlation matrices $\mathbf{D}_{i,j}$, N_m elements with the largest modulus is selected with indices written as $\left(i^{(k)}, j^{(k)}, \nu^{(k)}, \tau_2^{(k)}, \tau_3^{(k)}\right)$. The N_m candidate sequences are obtained based on the selected indices corresponding to outer and inner sequences, i.e., $i^{(k)}$ and $j^{(k)}$, respectively. The k-th KPS sequence relevant to k-th element $D_{i^{(k)},j^{(k)}}[\nu^{(k)}, \tau_2^{(k)}, \tau_3^{(k)}]$ can be expressed as

$$\mathbf{s}_k = \mathbf{p}_{i^{(k)}} \otimes \mathbf{q}_{j^{(k)}}. \tag{23}$$

Stage II: Fine Detection with Multiple Candidates: A more precise correlation with the received signal, as well as a finer Doppler compensation procedure is performed on the aforementioned N_m candidate sequences to further enhance the accuracy of detection.

Specifically, each candidate sequence is cyclically shifted based on its corresponding total delay $\tau_3^{(k)} L_I + \tau_2^{(k)}$ estimated in Stage I, i.e.,

$$\hat{s}_k[n] = s_k \left[\left\langle n - \left(\tau_3^{(k)} L_I + \tau_2^{(k)} \right) \right\rangle_L \right], \tag{24}$$

where $k = 1, 2, \cdots, N_m$. We obtain the cross-correlations between the received sequence and N_m cyclic shifted sequences $(\hat{\mathbf{s}}_1, \hat{\mathbf{s}}_2, \cdots, \hat{\mathbf{s}}_{N_m})$ to derive the finer delay estimation. The Doppler-compensated correlation can be written as

$$\Theta_k[\alpha] = \sum_{n=0}^{L-1} e^{-\mathrm{j}2\pi \frac{\alpha n}{N'_{\text{dop}}}} \hat{s}_k[n] \hat{y}[n], \tag{25}$$

where the normalized Doppler shift of each symbol is compensated by $\alpha \in [0, N'_{\text{dop}} - 1]$, and $\hat{y}[n]$ denotes the received preamble sequence represented in the time domain, i.e.,

$$\hat{y}[n] = \sum_{k=0}^{L-1} \hat{Y}[k] e^{\mathrm{j}2\pi \frac{nk}{L}}, \ n = 0, \cdots, L-1. \tag{26}$$

The number of multiplication required in the computation of the correlation result $\mathbf{\Theta}_k \in \mathbb{C}^{N'_{\text{dop}} \times 1}$ corresponding to all N_m candidate sequences amounts to $N_m(N'_{\text{dop}} \log N'_{\text{dop}} + L)$. Fig. 4 illustrates the complete procedure of the proposed two-stage low-complexity algorithm, and the total computational complexity required in the proposed detection algorithm can be written as

$$\begin{aligned} T_m =& N_I \left(N_c \log N_c + L \right) + NLN_{\text{dop}} \log N_{\text{dop}} \\ &+ N_m (N'_{\text{dop}} \log N'_{\text{dop}} + L). \end{aligned} \tag{27}$$

If the modulus of the k_0-th correlation result with normalized Doppler shift α_0 surpasses the test threshold γ, i.e.,

$$|\Theta_{k_0}[\alpha_0]| > \gamma, \tag{28}$$

we determine that there is an active UE whose preamble is based on the KPS sequence $\mathbf{s}_{k_0}$ with inner, outer, and total delay shifts presented as $\tau_2^{(k_0)}$, $\tau_3^{(k_0)}$ and $\tau_3^{(k_0)} L_I + \tau_2^{(k_0)}$, respectively. Substituting $\delta = \lfloor \frac{\tau_3^{(k_0)} L_I + \tau_2^{(k_0)}}{N_{\text{cs}}} \rfloor$ in (13), the preamble allocated for this active UE can be expressed as

$$\hat{s}_{k_0}[n] = s_{k_0} \left[n - N_{\text{cs}} \left\lfloor \frac{\tau_3^{(k_0)} L_I + \tau_2^{(k_0)}}{N_{\text{cs}}} \right\rfloor \right], \tag{29}$$

which can be directly utilized to infer the UE identity due to the one-to-one correspondence between each cyclically shifted version of the preamble and its associated UE. Furthermore, we can obtain the delay between the UE and BS as

$$\hat{\tau} = \tau_3^{(k_0)} L_I + \tau_2^{(k_0)} - N_{\text{cs}} \left\lfloor \frac{\tau_3^{(k_0)} L_I + \tau_2^{(k_0)}}{N_{\text{cs}}} \right\rfloor. \tag{30}$$

V. Numerical Results

The random access scenario is considered, where m-ZC and m-All KPS sequences are utilized as PRACH preambles. The preamble transmission is implemented using a DFT-s-OFDM waveform with 2048 subcarriers, carrier frequency of 6.7 GHz, and subcarrier spacing of 1.25 KHz. The channel model follows CDL-C, which is a representative propagation model specified in the 3GPP [14]. The system is configured with a single transmit antenna and four receive antennas, under a Doppler shift range of $[-5, 5]$ KHz. To validate the advantages of the proposed algorithm, its computational complexity and detection performance are compared with those of the conventional scheme.

The proposed method is assessed under two configurations, one of which involves only the first stage and and the other

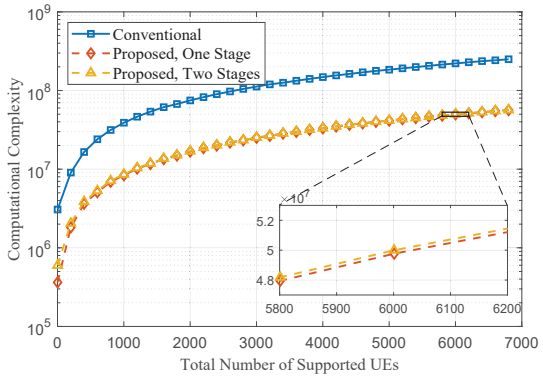


Fig. 5. Computational complexity comparison between the conventional and our proposed algorithms against total number of supported UEs.

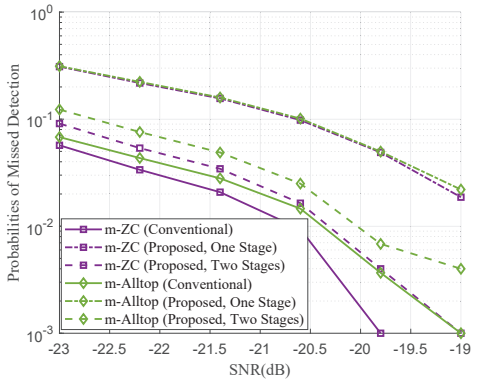


Fig. 6. Detection performance comparison between the conventional and our proposed algorithms.

incorporates both stages, denoted as "One Stage" and "Two Stages", respectively. For the "Two Stages" setup, the number of candidate sequences N_m is set to 10.

Figure 5 illustrates the computational complexity as a function of the number of supported UEs, quantified by the number of multiplications required. The conventional algorithm and the proposed Stage I employ $N_{\text{dop}} = 64$ Doppler compensation points, while the proposed Stage II uses $N'_{\text{dop}} = 2048$ points. As the number of UEs increases, the conventional algorithm experiences a rapid growth in computational load, since the received signal must be correlated with each sequence in the preamble set sequentially. In contrast, the proposed method (One Stage) decouples inner-layer and outer-layer correlations and simplifies the Doppler compensation process, leading to a notable reduction in computational burden. Furthermore, the incorporation of Stage II contributes only a slight increase in computational demand, thereby maintaining low overall complexity. For a system supporting 6400 UEs, the proposed algorithm (Two Stages) achieves an overall complexity reduction of 80% compared with the conventional approach.

In Fig. 6, the detection performance of the conventional and proposed algorithms is evaluated. The missed detection probability is plotted against SNR, while the false-alarm probability is fixed at 10^{-3} by appropriately tuning the detection threshold γ. As it can be seen in Fig. 6, the proposed algorithm (One Stage) exhibits a slight performance loss due to its simplified processing framework. However, by incorporating the fine detection stage, a marginal computational overhead in Stage II leads to a great enhancement in detection performance, which closely approaches that of the conventional algorithm.

VI. Conclusion

KPS sequence is proposed for 6G wireless systems, which includes an inherent segment structure to achieve large capacity and low-complexity detection simultaneously. Leveraging the structural property, the two-stage low-complexity detection algorithm is derived to achieve superior detection performance with relatively low implementation complexity. Simulation results validate that the proposed KPS sequence, in conjunction with the tailored detection method, could achieve extremely low computational complexity with negligible performance loss compared to its conventional counterpart under high Doppler shifts.

Acknowledgment

This work was supported by the National Natural Science Foundation of China under Grant 62471275. *(Corresponding author: Zhaocheng Wang.)*

References

[1] W. Saad, M. Bennis, and M. Chen, "A vision of 6G wireless systems: Applications, trends, technologies, and open research problems," *IEEE Netw.*, vol. 34, no. 3, pp. 134—142, May 2020.

[2] S. Li, F. Zhang, T. Mao, R. Na, Z. Wang, and G. K. Karagiannidis, "Transmit beamforming design for ISAC with stacked intelligent metasurfaces," *IEEE Trans. Veh. Technol.*, vol. 74, no. 4, pp. 6767–6772, Apr. 2025.

[3] N. Jiang, Y. Deng, A. Nallanathan, X. Kang, and T. Q. S. Quek, "Analyzing random access collisions in massive IoT networks," *IEEE Trans. Wireless Commun.*, vol. 17, no. 10, pp. 6853–6870, Oct. 2018.

[4] Y. Feng, *et al.*, "Modern random access-based IoT networks design: From a timeliness-fidelity perspective,"*IEEE Commun. Mag.*, vol. 62, no. 7, pp. 74–80, Jul. 2024.

[5] Z. Zhang and R. Rathi, "Novel access methods in cellular networks based on the analog bloom filter," *IEEE Trans. Wireless Commun.*, vol. 22, no. 8, pp. 5628–5641, Aug. 2023.

[6] T. Chiba, M. Sawahashi, Y. Kishiyama, and S. Suyama, "Performance of NR physical random access channel in the presence of carrier frequency offset in millimeter-wave bands,"*IEICE Trans. Commun.*, vol. E108-B, no. 2, pp. 164–177, Feb. 2025.

[7] F. Zhang, T. Mao, and Z. Wang, "Doppler-resilient design of CAZAC sequences for mmWave/THz sensing applications," *IEEE Trans. Veh. Technol.*, vol. 73, no. 2, pp. 2878–2883, Feb. 2024.

[8] M. H. AlSharif, M. Saad, M. Siala, M. Ahmed, and T. Y. Al-Naffouri, "Range estimation of a moving target using ultrasound differential Zadoff–Chu codes," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–15, 2021.

[9] 3GPP, "TR 38.211; Physical channels and modulation (Release 17)," 2022. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.211.

[10] R. -A. Pitaval, B. M. Popović, P. Wang, and F. Berggren, "Overcoming 5G PRACH capacity shortfall: Supersets of Zadoff–Chu sequences with low-correlation zone," *IEEE Trans. Commun.*, vol. 68, no. 9, pp. 5673–5688, Sep. 2020.

[11] S. W. Golomb, and G. Gong, *Signal Design for Good Correlation: For Wireless Communication, Cryptography, and Radar.* Cambridge, U.K.: Cambridge Univ. Press, 2005.

[12] R. W. Heath, T. Strohmer, and A. J. Paulraj, "On quasi-orthogonal signatures for CDMA systems," *IEEE Trans. Inf. Theory*, vol. 52, no. 3, pp. 1217—1226, Mar. 2006.

[13] L. Zhen, S. Li, Y. Wang, J. Su, and K. Yuyz, "PRACH sequence design for NR unlicensed spectrum," *Proc. IEEE Globecom Workshops*, (Rio de Janeiro, Brazil), Dec. 2022, pp. 1279–1284.

[14] 3GPP, "TR 38.811; Study on New Radio (NR) to support non-terrestrial networks (Release 15)," 2020. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.811.

[15] X. Lin, A. Adhikary, and Y. -P. Eric Wang, "Random access preamble design and detection for 3GPP narrowband IoT systems," *IEEE Wireless Commun. Lett.*, vol. 5, no. 6, pp. 640–643, Dec. 2016,